Appendix D
Auxiliary Display

Display page (F8) holds an Auxiliary Display with all variables necessary for process simulation and control. It also shows those control variables, normally not shown on real industrial control displays, that are helpful in the understanding of process control. This display shows variables such as Simulation Gain and values, Noise Gain and values, Disturbance Gain and values, special controller output and limit calculations etc.

Controller Variables

OP Controller output in % of range.

PV Process Variable in % of range.

SP Setpoint in % of range.

CSP Internal, computed setpoint in % of range.

CVPD CVPD is the increment value that occurs as a result of Proportional and Derivative control calculation. CVPD is used to increment (or decrement) all controller outputs by its value.

CVI CVI is the increment value that occurs as a result of Integral control calculation. CVI is used to increment (or decrement) all controller outputs by its value.

OPCALC. Status variable to define the type of OP-Limit calculation. OPCALC can be set to the status VIRTUAL or REAL. VIRTUAL permits the internal output value (OPVIRT) to assume values beyond any OP-Limit. The real output OP is then limited by OP-Limits.

Any internal controller calculation makes use of the virtual, internal and unlimited output value. This results in a saturated OP limit calculation. REAL causes both to be limited by OP-Limits, the internal output value (OPVIRT) and the true OP. This results in a NON-saturated OP limit calculation.

ACTION Direction of output control action. ACTION can be set to REVERSE or DIRECT. With DIRECT control action, the output value moves in the same direction as the PV value. With REVERSE control action, the output value moves in opposite direction to the PV value.

INIT This status variable shows whether a controller is initialized by another downstream controller.

MODE Status variable to define the MODE of operation of a controller. MODE can assume the status MANUAL, AUTO, CASC, I-MAN, I-AUTO or I-CASC. MANUAL mode disables automatic control of OP and enables the operator to manipulate OP manually. AUTO mode enables automatic control of OP and manual setting of SPE. CASC (cascade) mode enables automatic control, as in AUTO mode, but no operator setting of SPE is possible if another controller drives SPE as its primary controller. If two conditions in a cascade control system are true, the primary controller assumes the MODE I-MAN, I-AUTO or I-CASC. These two conditions are: the secondary controller is not in CASC mode and the secondary controller is configured to initialize the primary controller.
In this situation, the secondary controller's SPE is used to set the primary controller's OP. The primary controller waits until the secondary controller has been switched into CASC mode. Then, the primary controller changes from I-MAN to MANUAL or from I-AUTO to AUTO or from I-CASC to CASC. The mode indicated after I, as in I-AUTO, is called pending mode. For example, I-AUTO means the primary controller is presently initialized with AUTO mode pending.

CONFIG Status variable, defining Initialization and / or PV-Tracking.

INIT initializes an upstream primary controller's OP if the initializing controller (secondary) is not in CASC mode.
TRACKING forces SPE to assume (track) the value of PVE if the controller is not in AUTO or CASC mode.
I & TR performs both Initialization and PV-Tracking.

EQUATION Status variable, defining different selections of inputs for PID control.
TYPE A calculates PID-Control on ERR (ERR = PVE-SPE).
TYPE B calculates PI-Control on ERR (ERR = PVE-SPE) and D-Control on PVE only.
TYPE C calculates I-Control on ERR (ERR = PVE-SPE) and PD-Control on PVE only.

Process Simulation Variables

The parameters used for process simulation are unique to the process simulated. Knowledge of the simulation is not required to complete the exercises in this documentation. It is however, recommended to study the **application PCF** file if a deeper understanding is desired.

As most simulations contain noise and disturbance, the following is a description of them, using their generic names.

K-NOISE Gain of the noise simulation. The values of a random number generator will be superimposed on the simulated process variable. The magnitude of the random numbers are re-scaled by K-NOISE.

K-DISTURB Gain of the automatic random disturbance generator. Internally, within the disturbance generator, a raw disturbance value (RAW-DIST) is calculated by incrementing random numbers (positive and negative). This raw value has to go through a low-pass filter with the time constant TC-DISTURB. The output of the low-pass filter is DISTURB.

TC-DISTURB This time constant controls the speed (frequency) behavior of DISTURB.

DISTURB Value of the process disturbance. For manual control of DISTURB as with step functions, set K-DISTURB to 0 and TC-DISTURB to 999.

DIST-HI Disturbance high limit. DISTURB will not exceed DIST-HI when driven by the automatic random disturbance generator. DIST-HI does not limit manual changes of DISTURB.

DIST-LO Disturbance low limit. DISTURB will not go below DIST-LO when driven by the automatic random disturbance generator. DIST-LO does not limit manual changes of DISTURB.

SUM-DIST The sum of disturbance and process simulation.